ACSize — Master Project Specification

Project: ACSize – Offline Room Cooling Capacity Estimator & AC Selection Assistant
Module: IT3133 P — Mobile Communication and Computing (Practical)
Assessment: ICA 03 — Individual React Native Project
Version: 1.2
Prepared: 3 November 2026
Submission deadline: 15 November 2026
Status: All seven screen designs, numerical dataset, calculation defaults, navigation stack and offline architecture selected; implementation handover preparation pending.

> **Purpose of this document:** The authoritative planning baseline for a separate Codex implementation chat. It preserves approved choices, proposed constants, requirements, limitations, open items and development priorities. This is a mobile practical assessment—not professional HVAC engineering software.

1. Project Overview

1.1 Description

ACSize is a fully offline React Native mobile application for estimating a room's approximate cooling-capacity requirement in BTU/h and choosing a suitable air-conditioning solution. Users enter room dimensions and conditions, indicate installation and running-cost preferences, and receive a recommended nominal capacity and equipment profile from an original local AC catalogue. The app estimates approximate energy use and additional electricity charges, and explains the recommendation. An optional **Room Lab** allows users to experiment with room conditions, change AC equipment, view responsive results and explore simple educational comfort indicators. Saved room profiles can be reopened, edited or deleted.

1.2 Primary objective

"What AC cooling solution would be most suitable for my room, given its physical conditions, my preferences and approximate electricity costs?"

The primary user-facing output is a **recommended solution**, not an isolated BTU calculator and not a simulation app. Avoid promising a universally "optimal" real-world system: the recommendation is the best match within the app's simplified assumptions and offline catalogue.

1.3 Priorities

1. Reliable working React Native application.
2. Complete lecturer-requirement coverage.
3. Attractive, coherent, beginner-friendly UI.
4. Meaningful reactive interactions and navigation.
5. Local dataset, retrieval, search, filtering and details.
6. Reliable offline persistence.
7. Internally consistent, plausible approximate calculations.
8. Energy consumption and cost exploration.
9. Modest optional animations and visual polish.
10. Scientific precision is **not** a project priority.

The lecturer explicitly permits AI-assisted coding. The student must still understand the app’s navigation, state, storage, dataset and calculations.

2. Lecturer Requirements and Compliance

Lecturer requirement	Planned ACSize implementation	Status
Individual React Native application	Standalone React Native mobile app	Approved
Unique project title	ACSize title registered	User-confirmed complete
At least five screens/pages	Seven navigable routes	Approved direction
Suitable navigation	Primary tabs plus stack flows	Provisional mechanism
Original locally stored dataset	14 curated, generic AC profiles	Approved
Dynamic retrieval/display	Map structured records into catalogue and recommendations	Required
Search, filters, details	AC Library and AC Details	Approved
No internet/public APIs/online DB	Bundled catalogue and local storage	Approved
Individual GitHub repo; regular commits	Real commits and pushes through development	Required
Title registration by Nov 4, 2026	Registered	User-confirmed
Submission by Nov 15, 2026	Finished, tested app/repository	Pending

Do not assume there is a viva or a code-line demonstration: neither was established by the lecturer brief. Still prepare a short app walkthrough.

3. Scope and Boundaries

3.1 Supported rooms — approved

- Bedroom.
- Living room.
- Small office.

Not supported in v1: kitchens, large commercial/industrial spaces, multi-room system sizing or specialist cooling loads.

3.2 Core features — approved

- Three-step guided Room Setup with validation, defaults and units.
- Live area calculation and simple room preview.
- Approximate BTU/h estimation and transparent breakdown.
- Capacity-class recommendation and preference-based equipment selection.
- Fourteen-record offline AC catalogue and detail pages.
- Local searching/filtering and data-driven rendering.
- Approximate monthly AC kWh and additional energy charge.
- Room Lab with live condition and AC swapping controls.
- Educational, visual Comfort Explorer (no scientific sensation score).
- Save, reopen, edit and delete room profiles locally.

- Fully offline functionality.

3.3 Deliberate exclusions — deferred/out of scope

Cloud accounts, public APIs, manufacturer feeds, live prices, exact Sri Lankan bill prediction, professional heat-load/psychrometric models, PMV/PPD, cooldown-time simulation, AR/3D, drag-and-drop furniture, real HVAC installation design, multi-room systems, full simulation histories, advanced compressor cycling and installation-cost estimates. Window AC and evaporative cooler categories are excluded in v1.

3.4 Optional polish only after MVP

Subtle transitions, animated numbers, expanded AC comparisons, energy charts, more room templates and extra themes.

4. Screen Architecture and Navigation

4.1 Seven routes — approved

Route	Responsibility
Home	Brand dashboard, Start New Estimate, recent rooms, shortcuts
Room Setup	One route containing three guided steps
Recommendation	Best-match AC, cooling estimate, running cost, reasons, alternatives
Room Lab	Hybrid interactive dashboard and exploratory controls
AC Library	Local searchable/filterable AC catalogue
AC Details	Detail view for an AC record; optional Use This AC action
Saved Rooms	List/open/edit/delete saved configurations

Energy & Cost and **Comfort Explorer** are internal Room Lab sections, **not separate routes**. Results and AC Library must not become unrelated “journeys”: all of them support the room’s final recommendation.

4.2 Navigation — provisional implementation

Bottom navigation: **Home / AC Library / Saved Rooms**. Room Setup, Recommendation, Room Lab and AC Details are stack routes implemented with **React Navigation Native Stack + Bottom Tabs**. The catalogue operates in (1) free browsing mode from main navigation and (2) selection mode entered from Room Lab. In selection mode, **Use This AC** returns to Room Lab without losing the working scenario.

Main journey: **Home → Room Setup → Recommendation → Save Solution or Explore in Room Lab**. Saved rooms reopen into existing routes; no duplicate editing pages.

5. Room Setup — Three-Step Guided Form

Approved: One guided route with progress 1/3–3/3, Back/Next controls, sensible defaults, inline errors, keyboard handling, preserved values, hints and responsive room illustration. Do not produce final recommendation until **Find My AC** is tapped after the third step.

5.1 Step 1 — Your Room

Input	Control	Default	Notes
Name	Text	My Bedroom	Nonempty; suggested name changes when room type changes, without erasing intentionally edited text
Room type	Three cards	Bedroom	Bedroom, Living Room, Small Office
Length	Decimal	4 m	Suggested bounds 2–15 m
Width	Decimal	3 m	Suggested bounds 2–15 m
Height	Decimal	2.7 m	Suggested bounds 2–5 m; affects simplified load
Units	Toggle	Metric	Metric/imperial; convert values, never just relabel

Area = length(m) × width(m). Display in m² or ft² to match user units. **Normalize dimensions internally to meters** (recommended). 1 m = 3.28084 ft; 1 ft = 0.3048 m. Limits are proposed interface bounds, not building standards, and should be enforced equivalently in either unit system.

5.2 Step 2 — Room Conditions

Input	Values	Default	Control
Occupants	1–10	2	Stepper
Sunlight	Low / Medium / High	Medium	Segmented
Windows	None / Small / Large	Small	Segmented
Roof/ceiling	Concrete slab; sheet/tile without ceiling; sheet/tile with ceiling; another occupied floor above	Sheet/tile with ceiling	Choice list

Approved replacement: Use roof/ceiling configuration **instead of** the generic Poor/Average/Good insulation input; it is more understandable in the target context. Roof conditions are simplified heuristics and must not be represented as universally valid heat-performance rankings.

5.3 Step 3 — AC Preferences

Input	Values	Default
Installation	Split / Portable / Either	Either
Daily use	1–16 h/day	8 h
Main priority	Lower initial cost / Balanced / Lower electricity use	Balanced

No exact purchase budget: equipment profiles are generic and have cost *tiers*, not verified market prices. Electricity unit rate and any advanced assumptions belong in Room Lab's Energy section, not the initial questionnaire. Portable AC requires suitable exhaust outdoors.

5.4 Form interactions

- Next validates current step; Back preserves all values.
- Step 1 shows dynamic area; Step 2 can update illustration.

- Changing room type must not erase user dimensions.
- When editing a saved room, prepopulate all fields.
- Preserve in-progress scenario during ordinary navigation.
- Show clear inline validation and never allow NaN/negative dimensions to reach calculations.
- Buttons remain reachable when keyboard is visible.
- Primary final action: **Find My AC**.

6. Cooling-Capacity Estimation — Simplified v1 Proposal

Status: The basic heuristic structure is agreed. **Exact coefficients are provisional** pending final lock; do not describe them as an industry sizing standard.

6.1 Formula

```
AreaM2 = LengthM × WidthM
BaseLoadBTU = AreaM2 × 700
CombinedAdjustment = RoomTypeAdj + SunAdj + WindowAdj + RoofAdj + HeightAdj
ExtraOccupantBTU = max(0, Occupants - 2) × 500
RawBTU = BaseLoadBTU × (1 + CombinedAdjustment) + ExtraOccupantBTU
EstimatedBTU = round(RawBTU / 100) × 100
```

All percentage adjustments are **added**, not compounded. Two occupants are included in baseline; one occupant gets no negative credit. Round displayed output to nearest 100 BTU/h. Area coefficient is **700 BTU/h/m²** (application heuristic).

6.2 Coefficients

Factor	Setting	Additive adjustment
Room type	Bedroom	0%
	Living room	+4%
	Small office	+8%
Sunlight	Low	−5%
	Medium	0%
	High	+10%
Window category	None	−3%
	Small	0%
	Large	+8%
Roof	Another occupied floor above	0%
	Sheet/tile roof with ceiling	+3%
	Concrete slab directly exposed	+8%
	Sheet/tile roof without ceiling	+12%

Height adjustment: **(HeightM − 2.7) × 0.03**, i.e., 3 percentage points per meter difference versus 2.7 m. Extra occupants over two: **+500 BTU/h each**. These are intentionally modest project-specific assumptions; they are not externally validated engineering constants.

6.3 Worked reference case

Default bedroom: 4 m × 3 m × 2.7 m; two occupants; medium sun; small window; roof with ceiling.

```
Area = 12 m²
Base = 12 × 700 = 8,400 BTU/h
Adjustments = 0 + 0 + 0 + 0.03 + 0 = +3%
Extra occupants = 0
Raw = 8,400 × 1.03 = 8,652 BTU/h
Rounded = 8,700 BTU/h
Nominal split class = 9,000 BTU/h
```

Optional cooling-capacity conversion: **kW(cooling) = BTU/h × 0.000293071**. Show approximate labels, breakdown and assumptions rather than false precision.

7. AC Catalogue — Fourteen Generic Profiles

Approved: 12 split and 2 portable records in an original bundled local dataset. They are **illustrative equipment profiles**, not actual branded products or verified market specifications. Use dynamic rendering and shared schema, not fourteen handwritten cards.

7.1 Split profiles (12)

Proposed ID	Profile	Nominal BTU/h	Technology
SPL-09-STD	9K Standard	9,000	Fixed-speed split
SPL-09-INV	9K Inverter	9,000	Inverter split
SPL-09-ECO	9K Eco Inverter	9,000	Efficient inverter split
SPL-12-STD	12K Standard	12,000	Fixed-speed split
SPL-12-INV	12K Inverter	12,000	Inverter split
SPL-12-ECO	12K Eco Inverter	12,000	Efficient inverter split
SPL-18-STD	18K Standard	18,000	Fixed-speed split
SPL-18-INV	18K Inverter	18,000	Inverter split
SPL-18-ECO	18K Eco Inverter	18,000	Efficient inverter split
SPL-24-STD	24K Standard	24,000	Fixed-speed split
SPL-24-INV	24K Inverter	24,000	Inverter split
SPL-24-ECO	24K Eco Inverter	24,000	Efficient inverter split

For these simplified split examples, comparison capacity = nominal capacity.

7.2 Portable profiles (2)

Proposed ID	Profile	Nominal BTU/h	Illustrative comparison BTU/h
POR-09-CMP	Portable Compact	9,000	7,200
POR-12-PLUS	Portable Plus	12,000	9,600

The comparison values use an **approved illustrative 80% heuristic**, not an official SACC value or universal performance ratio. Portable ACs should be qualified as requiring outdoor heat exhaust. No broad portable range is planned because the extra product variety would add little assessment value.

7.3 Record schema

```
id, name, category, technology,
nominalCapacityBTU, comparisonCapacityBTU, coolingCapacityKW,
illustrativeInputPowerKW, purchaseCostTier, efficiencyTier,
features[], description, recommendedUse, installationNotes, limitations
```

Enumerations: category = split/portable; technology = fixed-speed/inverter/eco-inverter/portable; purchase tier = budget/mid-range/premium; efficiency tier = standard/efficient/high. Names/descriptions should be original curated project data.

7.4 Approved electrical input and purchase tiers

See **Section 28.1** for the complete approved 14-profile power/cost table. The 12K values are Standard 1.25 kW, Inverter 1.05 kW, Eco 0.90 kW; all other profile power values and tiers are likewise fixed in Section 28.1. Fictional explanatory descriptions/features are generated consistently from the schema, with no implied manufacturer verification.

7.5 Library behaviors

Search names/capacity/keywords; filter category (split/portable), nominal capacity, technology and/or efficiency; clear filters; dynamic cards; open details; useful empty state. Details include capacities, power, equipment and cost tiers, features, use cases, installation notes and limitations. Selection mode adds **Use This AC** and returns to active Room Lab scenario.

8. Recommendation Algorithm — Approved Rule-Based Design

1. Calculate estimated room BTU/h.
2. For each equipment record, determine **IsSufficient = comparisonCapacityBTU ≥ EstimatedBTU**.
3. Filter/prioritize by installation preference. “Either” prioritizes suitable split units; “Portable” prioritizes adequate portable options; “Split” prioritizes split.
4. Within the preferred suitable category, choose the **smallest sufficient comparison-capacity class**.
5. Within the chosen split capacity, match the main priority:
    - Lower initial cost → Standard.
    - Balanced → Inverter.
    - Lower electricity use → Eco Inverter.
6. For portable units, choose a sufficiently rated available profile; do not fabricate three technology tiers that do not exist in the portable dataset.
7. Produce dynamic reasons (“Meets estimated requirement,” “Matches installation preference,” “Prioritizes energy efficiency,” etc.) and alternative profiles.

Insufficiency: Never recommend an insufficient unit solely to satisfy installation preference. If no portable option meets demand, explain and offer suitable split alternatives. If none of the 14 records meets demand, show **“No suitable AC found in the current library.”**

Important distinct states: `recommendedACId` is the algorithm’s current best match; `selectedACId` is equipment the user is testing manually. Selecting a different AC must **not** silently rewrite the recommendation.

9. Energy and Electricity-Cost Estimates

9.1 Monthly electricity

```
MonthlyKWh = InputPowerKW × DailyHours × DaysPerMonth × AveragePowerFactor
DaysPerMonth = 30 (default)
AveragePowerFactor = 0.65 (approved fixed v1 value)
DailyHours = user value, default 8
```

Example: 1.05 kW × 8 h/day × 30 days × 0.65 = **163.8 kWh/month**.

The power factor here is **assumed average electrical input utilization**, not power-factor correction and not a scientifically predicted duty cycle. Real compressors, inverter controls and climate conditions vary.

9.2 Approximate cost

```
AdditionalEnergyChargeLKR = MonthlyKWh × UserRateLKRPerKWh
```

Currency: **LKR**. User can adjust the rate in Room Lab. Approved demonstration default = **Rs. 40/kWh**, user-adjustable; not an official utility tariff. Display **“estimated additional energy charge”** instead of promising an exact domestic electricity bill difference. Official tiered Sri Lankan tariff modeling is optional/deferred, not core v1.

9.3 Dependencies

- Changing AC profile updates profile input power, kWh and cost.
- Increasing hours increases kWh/cost.
- Increasing electricity rate increases cost but not kWh.
- Changing room conditions updates cooling load/recommendation, not a manually selected AC’s fixed stored electrical input.
- Comfort controls alone do not update AC consumption.

10. Recommendation Screen — Primary Result

Five vertically stacked sections:

1. **Best Match:** prominent profile, nominal BTU/h, technology, Best Match label.
2. **Cooling Estimate:** rounded room requirement, recommended nominal capacity, optional cooling kW, approximate suitability and brief breakdown.
3. **Running Cost:** illustrative kWh/month, additional energy charge and assumptions.
4. **Why This AC?:** two or three reasons dynamically tied to rules.
5. **Alternatives & Actions:** appropriate alternatives; **Explore in Room Lab**, **Save Solution**, optionally Edit Room and View Details.

Recommendation is fully usable **without opening Room Lab**. Keep this screen answer-first and not full of sliders. Provide clear no-match states and “approximate estimate” labeling.

11. Room Lab — Hybrid Interactive Dashboard

Approved: Room Lab is optional *exploration*, not the product's main purpose. It contains a compact room illustration and summary at the top with organized expandable sections beneath. Avoid a large permanently sticky illustration that takes over a phone screen.

11.1 Summary

Room name/type, compact visual, current BTU/h, recommended and selected AC, monthly energy summary; update relevant values immediately as scenario changes.

11.2 Room Conditions

Editable dimensions, height, occupants, sunlight, windows and roof. Reuse the same cooling-estimation function as Recommendation. Room illustration updates simply (roof/window/sun/occupant icons); avoid costly 3D/thermal animations.

11.3 AC Selection

Show automatic recommendation and currently selected profile separately. **Change AC** opens local library selection mode, then Details → **Use This AC** → Room Lab. Display suitability, nominal/comparison BTU/h, power and costs for selected equipment. A lightweight recommended-versus-selected comparison is enough; no separate comparison-history app.

11.4 Energy & Cost

Daily hours, editable example LKR/kWh tariff, illustrative electrical power, kWh/month, additional monthly energy charge, and simple comparison when testing another profile. Values react to controls.

11.5 Comfort Explorer — Option A approved

An **educational visual** explorer, not PMV/PPD or a cooldown simulation. Hypothetical indoor air temperature, relative humidity and airflow change a lightweight illustrative indicator and explanation. Proposed ranges: temperature **20–34 °C**, humidity **30–90%**, airflow **Still / Gentle / Strong**. Potential labels: **Cool / Comfortable / Warm**. The precise mapping/thresholds remain **open**; do not make up a scientific "feels like 29°C" prediction or pretend to measure the room. Changing occupancy/roof does not automatically predict a measured indoor temperature.

11.6 Baseline and actions

Room Lab operates on a working copy/snapshot. **Reset** restores values to the baseline at entry. **Apply Changes** adopts current scenario and updates Recommendation. **Save Solution** persists new/updated room intentionally. No silent overwrite on sliders, no complex undo history.

12. Saved Rooms and Local Storage

12.1 Suggested room schema

```
id, name, roomType,
lengthM, widthM, heightM, preferredUnit,
occupants, sunlight, windowCategory, roofConfiguration,
installationPreference, dailyUsageHours, selectionPriority,
selectedACId, lastRecommendedACId,
createdAt, updatedAt
```

Derived results can be recomputed from inputs and catalogue; optionally store a display snapshot with method/data version so stale results can be identified. Exact snapshot policy remains open.

12.2 Required persistence

Create/list/open/edit/delete; survive app restart; confirmation before deletion; helpful empty state with **Create Room**. Cards show name/type/area/last approximate BTU/h/recommended nominal capacity/updated date. Room edits repopulate guided setup, retaining all input values.

Provisional technology: bundled TypeScript/JSON for reference catalogue; AsyncStorage for user records and settings. SQLite is unnecessary for this scope unless class lab restrictions require otherwise.

13. Visual Design and UI Principles

Approved direction and first three screen designs: light, airy, modern **climate-tech** look. See Sections 21–24 for approved implementation-level visual decisions. Proposed colors:

Role	Hex
Navy text	`#0F172A`
Primary teal	`#0D9488`
Light cyan	`#38BDF8`
Pale blue	`#E0F2FE`
Background	`#F8FAFC`
Warm sun/warning accent	`#F59E0B`

Rounded cards, generous spacing, consistent icons, readable typography, large taps and clear errors. Cooling-related colors should not be the only source of meaning; pair colors with labels. Room illustration can be 2D or pseudo-isometric with simple variations for room type, roof, window, sun and occupant state. Use subtle fades and number updates if inexpensive.

Visual hierarchy: recommended AC first → estimated room BTU/h → estimated consumption/cost → reasons and alternatives → optional Room Lab actions. The first impression should be an AC selection assistant, not a physics toy.

14. Technical Architecture — Approved Stack

Selected: React Native with **Expo + TypeScript**, **React Navigation Native Stack + Bottom Tabs**, **AsyncStorage**, React Context/hooks, locally bundled 14-profile catalogue, pure calculation modules, React Native StyleSheet/theme and local SVG/vector illustrations. The previous Expo Router suggestion is rejected for v1. See **Section 29** for the complete navigation, state, persistence and file-organization contracts.

Conceptual modules: app shell/navigation/theme, room setup/saved rooms, catalogue, cooling engine, recommendation engine, Room Lab scenario, energy and comfort calculations, and persistence/error handling. Do not duplicate formulas across screens; do not rely on network assets/services. Important error cases: invalid dimensions, no sufficient AC, corrupted saved data, missing profile ID, storage failures, empty search and leaving modified Room Lab.

15. Behavior Invariants and Tests

15.1 Calculation consistency

- More floor area must increase cooling requirement with conditions held constant.
- More ceiling height must not lower load under v1 heuristic.
- More occupants over baseline must not lower load.
- Higher sunlight must not lower load.
- Roof/window selection uses the correct proposed coefficient.
- More operating hours increases consumption and cost.
- Higher electricity rate increases cost only.
- Lower electrical input (same use assumptions) lowers modeled kWh.
- Manually changing selected AC does not secretly change recommended AC.
- Under-capacity profile must never be labeled a guaranteed sufficient match.
- Calculation output agrees across Recommendation and Room Lab.

15.2 UI/navigation/storage acceptance

- ☐ Seven navigable routes work correctly.
- ☐ Guided form step/back/next and preservation work.
- ☐ Unit conversions and validation work.
- ☐ Default bedroom example produces **8,700 BTU/h** under provisional formula.
- ☐ Recommendation follows category, capacity and priority rules.
- ☐ Portable insufficient/no-match states are understandable.
- ☐ Fourteen local records display dynamically.
- ☐ Search, filters and details work without internet.
- ☐ Choosing equipment in library returns to active Room Lab.
- ☐ Room Lab Reset/Apply behaves correctly.
- ☐ Comfort controls update educational visuals without invented temperature prediction.
- ☐ Energy/cost controls update values consistently.
- ☐ Create/edit/delete persisted rooms; survive app restart.
- ☐ Good empty/error states, keyboard behavior and mobile-size layout.
- ☐ Full airplane-mode test passes.
- ☐ GitHub repository contains all source and genuine regular commits.

Suggested worked scenarios: default bedroom, large living room, small office, high sun, each roof condition, higher ceiling, >2 occupants, insufficient portable options, beyond 24K catalogue capacity, input-power comparisons and differing usage/rates.

16. Development and Delivery Plan

Date target	Work
Nov 3–4	Consolidate planning; registration confirmed
Nov 4–5	Finalize visuals, remaining coefficients/data values and architecture
Nov 6–11	Codex implementation in testable milestones
Nov 12–14	Offline QA, fixes, polish, documentation and GitHub check
Nov 15	Submission and final verification

Suggested implementation order: 1. Project initialization and navigation/theme. 2. Local catalogue and shared data types. 3. Guided Room Setup. 4. Cooling estimator and tests. 5. Recommendation engine/result UI. 6. Saved rooms persistence. 7. Searchable AC Library and Details. 8. Room Lab integration and AC selection. 9. Energy calculator and comfort explorer. 10. Illustrations/polish, end-to-end offline tests and repository documentation.

Implementation is for a **separate Codex chat**, not this planning conversation. Do not attempt to build every feature in one unverified step. Explain code/navigation/storage/calculations to the student as milestones are finished.

17. Approved Decision Register

- **Concept/title:** ACSize Offline Room Cooling Capacity Estimator & AC Selection Assistant — approved and registered.
- **Audience/scope:** bedrooms, living rooms and small offices — approved.
- **Assessment emphasis:** React Native functionality and visual quality; plausible approximate estimates — approved.
- **Catalogue:** 12 split + 2 portable generic offline profiles — approved.
- **Recommendation:** rule-based capacity-first, then installation/priority matching — approved.
- **Setup:** three-step guided flow, metric default with imperial support — approved.
- **Room factors:** occupants, sunlight, window category, roof/ceiling; retain ceiling height; remove generic insulation rating — approved.
- **Primary output:** full recommended solution with explanation and cost; not a simulation-first app — approved.
- **Room Lab:** optional hybrid dashboard, baseline/reset/apply, current-selected versus recommended AC — approved.
- **Comfort:** educational and visual Option A only — approved.
- **Energy:** simple kWh and adjustable approximate rate, no exact bill claim — approved direction.
- **Storage and required browsing:** persistent saved rooms, dynamic local catalogue with search/filter/detail — approved.
- **Seven routes:** approved direction; implementation routing library pending.
- **Visual identity:** modern climate-tech light design approved; Home, Room Setup and Recommendation layouts approved (Sections 21–23); Room Lab/Library/Details/Saved visual layouts pending.
- **Workflow:** continue planning here, then build through Codex with a finalized spec — approved.

18. Resolved and Remaining Decisions — Version 1.2

Most of the v1.1 open decisions are now resolved. The **current authoritative decisions** are recorded in **Sections 26–31**. The old v1.1 open-item table is preserved in the separately issued v1.1 document for historical reference rather than repeated here as if still pending.

Resolved in v1.2: Fourteen complete AC numeric profiles, cooling coefficients, portable capacity assumption, energy defaults (30 days/0.65/Rs.40), Comfort Explorer classification, all seven visual designs, selection/save lifecycle, React Navigation rather than Expo Router, and AsyncStorage/React Context architecture.

Still open for implementation detail: Final copy/tags for 14 generic entries, exact Expo-compatible dependency versions, optional custom electricity-rate text entry, minor UI microcopy, Android build/test arrangement, and the purely presentational threshold for *Higher Than Estimated Need*. No major feature or app-scope decision remains undecided.

19. Scope-Change and Version Control Policy

The master specification is the single planning reference. For each proposed addition, evaluate: (1) relevance to AC sizing/selection, (2) usefulness for the React Native assessment, (3) genuine user-facing improvement, (4) complexity, (5) November 15 schedule risk, and (6) conflicts with approved behavior. Do not accept every proposal automatically. When an addition is approved, update affected sections, data contracts, tests, decision register and version number. Clearly note what changed. Only update this specification when the student requests an update.

20. One-Paragraph Product Experience

The user opens an attractive offline AC sizing assistant, creates a bedroom, living room or office in a guided three-step form, chooses basic conditions and AC preferences, and taps **Find My AC**. ACSize calculates approximate cooling demand, recommends a suitable generic AC profile from its local catalogue, explains the match, shows illustrative monthly energy consumption and additional cost, and offers alternatives. The user can save the result or explore the room in a hybrid Room Lab, where changing conditions, equipment and usage values updates the relevant output; educational comfort controls visually demonstrate hypothetical temperature/humidity/airflow conditions without claiming real temperature prediction. Saved rooms and catalogue browsing work without internet. **The app is a practical selection assistant first, an interactive learning experience second.**

21. Visual Design Approval — Modern Climate-Tech (Approved)

This section records decisions finalized **after** specification v1.0. It takes precedence over earlier descriptions that say visual designs are pending, but does not alter technical assumptions or catalogue values.

- **Theme:** Modern Climate-Tech **light mode**; calm white/off-white screens, teal/cyan cooling accents, navy typography and restrained warm amber for sunlight or caution. Do not substitute a dark futuristic theme.
- **Palette baseline:** Background `#F8FAFC`; teal `#0D9488`; navy `#0F172A`; cyan `#38BDF8`; pale blue `#E0F2FE`; amber `#F59E0B`. Hero surface may use light mint/cyan (concept mockup `#E5F7F8`). Exact shades remain adjustable for contrast.
- **Components:** rounded, generous-spaced cards, clear selected states, legible labels, consistent icons, measured visual hierarchy; avoid animation-heavy graphics, unnecessary live-weather displays and decorative metrics.
- **Layout targets (not fixed device coordinates):** horizontal padding around **20 px**; standard card radius **16 px**; hero radius **24 px**; primary button height around **52 px**; common spacing **12–20 px**; card padding **16 px**; headings **25–28 px**; body **14–16 px**. Use safe areas and scrolling for smaller screens.

- **Brand:** ACSize wordmark plus simple snowflake/airflow icon usable in header, splash and app icon. The exact vector logo artwork is not yet fixed.
- **Reusable art direction:** lightweight pseudo-isometric/2D room graphic reused across Home, Room Setup and Room Lab; it conveys room identity, not engineering-accurate geometry. Sunlight/heat may use amber, cooling teal/cyan. Prefer bundled vector/illustration assets and no network-loaded images.
- **Accessibility:** labels must accompany visual/color status; readable text, clear errors, sensible touch targets; don't rely on color alone.

22. Home Dashboard — Approved Visual and Behavior Specification

Primary job: make **Start New Estimate** the clear main action, explain ACSize in seconds, and provide quick entry to the catalogue and saved rooms. Do **not** show meaningless pre-estimate BTU gauges, fake live temperatures, usage charts, online weather or statistics.

22.1 Approved structure (top to bottom)

1. **Brand header:** snowflake/airflow logo, **ACSize**, short tagline (concept: **"Smart cooling starts here"**); right-side small info icon.
2. **Hero card:** broad pale mint/cyan rounded surface, eyebrow label **"YOUR COOLING ASSISTANT"**, headline **"Find the right AC for your room."**, supporting line **"Estimate cooling needs, discover suitable AC options and explore running costs."**
3. **Hero illustration:** a compact illustrated bedroom featuring wall-mounted AC, window, bed, subtle airflow and mild sunlight. Static/bundled, not a complex dynamic scene; enough negative space for text. Must scale on small devices.
4. **Main teal CTA: Start New Estimate** with right arrow, inside hero. Caption: **"3 simple steps · Works completely offline"**.
5. **Quick Access:** two clear cards: **AC Library** ("Browse 14 AC profiles") and **Saved Rooms** ("Revisit your solutions").
6. **Secondary context:** one small **Cooling tip** card, using locally stored original explanations. Prefer one selected tip per app launch over a timer-driven carousel.
7. **Primary bottom navigation: Home · Library · Saved** with current tab indicated by both text and icon/color, respecting device safe area.

22.2 User states and actions

- **First launch/zero saves:** show hero, quick access and tip; no empty dummy recent-room statistics.
- **With saved rooms:** add or substitute a compact **Continue Last Room** card for the latest saved room; opens stored room and recommendation, without creating a duplicate.
- **Start New Estimate:** creates a *fresh unsaved working scenario* with the approved defaults and opens Room Setup step 1; never overwrites a stored room.
- **AC Library shortcut:** opens general catalogue **browsing mode** without needing an active room.
- **Saved Rooms shortcut:** opens saved list.
- **Information icon:** opens a small modal/panel with purpose, offline nature, and an educational approximate-estimate disclaimer. No additional route is required.
- **Tips:** sourced locally, static per appearance/session; no online lookups.

22.3 Explicit exclusions

No Home cooling-gauge without a configured room; no live weather; no temperature sensor UI; no animated heat particles, automatic switching carousels or unrelated analytics widgets.

23. Room Setup — Approved UI and Interaction Specification

The **three-step guided wizard is one route**. Modern Climate-Tech styling, white cards, teal selected states, short helper copy and a compact three-segment progress indicator. Back/Continue actions remain reachable in a safe-area-aware bottom action region with keyboard handling.

23.1 Universal layout and navigation

- Top: Back arrow to previous app screen where appropriate, title **“Set Up Your Room”**, **“1 of 3” / “2 of 3” / “3 of 3”** and three horizontal progress segments.
- Main step content scrolls independently if needed; smooth short slide/fade step transition is optional. Do not implement elaborate wizard animation.
- Buttons: **Continue** on steps 1 and 2; **Back** on steps 2 and 3; **Find My AC** on step 3.
- Persist all form inputs across steps. Opening an existing saved room for editing must restore values; abandoning an unsaved scenario must not modify saved data.
- Validate after a field is touched or on Continue, rather than showing errors immediately in untouched defaults.
- Normalize length/width/height internally to meters; metric is default, imperial is a true conversion, not relabeling. Existing numeric limits from Section 5 remain provisional until final lock.
- Use numeric-friendly keyboards; maintain primary actions above the on-screen keyboard or provide keyboard-aware scrolling.

23.2 Step 1 — Your Room (approved visual layout)

1. Heading **“Let’s start with your space”** and brief description.
2. Three selectable room-type cards with icons: **Bedroom** (bed), **Living room** (sofa), **Small office** (desk/monitor). Selected type outlined or softly filled teal.
3. **Small 2D/pseudo-isometric preview** immediately below type cards. Swap simple furniture/icon arrangement for bedroom, living and office. It is illustrative and **must not require floor geometry/furniture to scale to entered dimensions**.
4. Room name field; suggested names **My Bedroom**, **My Living Room**, **My Office**.
5. **Length / Width / Height** numeric inputs in a compact layout plus **Meters / Feet** selector; dynamic calculation of area in displayed units.
6. Light-tinted **Calculated floor area** summary card.

Important correction to prior prototypes: when changing room type, auto-update room name *only when the current name is still the automatically generated default*; preserve any user-customized name such as “Dad’s Room.” Dimensions must also remain intact. For editing, don’t replace an existing user name.

Validation: nonempty name; positive decimal dimensions within app bounds; inline field-level messaging, no progression while invalid. Preview still displays even if a numeric value is incomplete.

23.3 Step 2 — Conditions (approved visual layout)

1. Heading **“Describe the room conditions”**, brief explanation.
2. **Occupants** numeric stepper with plus/minus, 1–10, default 2.
3. **Sunlight** segmented Low / Medium / High, default Medium, with contextual explanation.
4. **Window category** segmented None / Small / Large, default Small.
5. **Roof & ceiling** selectable list/dropdown for Sheet/tile roof with ceiling (default), Concrete slab, Roof without ceiling, Another occupied floor above; concise helper text.
6. Compact **Conditions summary** showing selected characteristics.

No repeated large hero illustration on Step 2; focus on efficient data entry and accessible controls. The roof input replaces generalized insulation (already established in Section 5).

23.4 Step 3 — AC Preferences (approved visual layout)

1. Heading **“What matters most to you?”** and explanation that preferences influence the chosen equipment, not physical room area.
2. Installation preference radio group **No preference / Permanent split / Portable** (default No preference).
3. Daily usage slider, **1–16 hours**, default 8.
4. Priority radio group **Lower purchase cost / Balanced cost and efficiency / Lower electricity consumption** (default Balanced).
5. Compact **Ready to find your AC** summary with room name, usage and preference.
6. Distinct primary **Find My AC** button launching Recommendation after validating all setup data.

Don’t add a purchase-budget input, tariff table, humidity or outdoor-temperature questionnaire to the three-step flow.

23.5 Visual polish and interaction quality

- Contextual info icons for roof, installation and technology where helpful.
- Selected states use both border/filled treatment and text/icon.
- Step contents need no separate navigation routes.
- Light transitions optional and subordinate to functional reliability.
- Recommendation is shown only after Find My AC; Room Lab handles subsequent live updates.

24. Recommendation Screen — Approved UI and Behavior Specification

The **Recommendation screen is the product’s primary payoff**, not a calculator configuration page. Scroll vertically with the best match near the top, substantial whitespace and concise explanations. All values are derived from the same room state, dataset and estimate engine; illustrative values below are not hardcoded UI outputs.

24.1 Layout hierarchy

1. **Heading:** small eyebrow **“ACSIZE / ROOM ANALYSIS”**, title **“Your AC Solution”**, and room name with floor area. Small award icon permissible.
2. **Best Match card:** largest teal/mint-tinted feature card, badge **“Best Match”**, generic locally bundled AC illustration, recommended profile name, nominal BTU/h, technology/category and short room-specific explanation. Keep the recommended solution visually dominant.
3. **Cooling Capacity Match:** clear required BTU/h versus chosen AC **comparison capacity**. Use a **horizontal progress/comparison bar**, not a large speedometer/dial. Show both numeric values and a textual suitability state.
4. **Estimated Electricity Usage:** two compact cards for **monthly kWh** and **approximate additional monthly energy charge**, plus stated usage/rate assumptions; do not show fabricated values when a profile has missing power data.
5. **Why ACSize recommends this:** 2–3 dynamically generated check-mark reasons tied to the recommendation algorithm, capacity sufficiency, installation choice and priority.
6. **Other Suitable Options:** compact alternative profile cards (e.g., lower initial-cost/energy use options), with entry to AC Details or Room Lab; avoid claiming insufficient alternatives are suitable.
7. **Actions:** high-priority **Explore in Room Lab**, outlined **Save Solution**, and accessible **Edit Room**; a small collapsible **“How was this calculated?”** section containing base, percentage/condition adjustments, occupancy and rounded result; approximate-estimate note near bottom.

24.2 Capacity bar states

- **Suitable:** load within AC's applicable comparison capacity; teal and verbal label **"Suitable"**.
- **Insufficient:** selected AC comparison capacity below estimated load; amber and text **"Below Estimated Requirement"**. Do not draw misleading overflowing percentage bars; clamp graphic while showing raw values.
- **Higher capacity:** notably larger than need, label **"Higher Than Estimated Need"** rather than "better/excellent." Exact threshold for "notably" is pending final UI logic.
- Portable profile comparisons must use **comparisonCapacityBTU**, while visible nominal BTU/h remains clearly distinguishable.

24.3 Explanation and example consistency

For default 4m × 3m bedroom, 2.7m high, medium sunlight, small windows and roof with ceiling, the v1 proposed heuristic yields **8,700 BTU/h** and a **9K nominal split class**. Explainable breakdown: base **8,400 BTU/h**, combined roof/conditions **+252 BTU/h** raw, occupants **0**, rounded **8,700 BTU/h**. Display consistent rounded values, avoid conflicting totals caused by independently rounded components.

Electricity summary for this 9K profile remains **unpopulated as a fixed value until the 9K profile's input power is finalized**. It should be computed when a record exists; the demonstration mockup used an intentional placeholder.

24.4 Actions and states

- **Explore in Room Lab:** open a working scenario using Recommendation as baseline and the recommended AC as initial selected equipment. Keep stored recommendation and manual selection distinct.
- **Save Solution:** explicit create/update of a stored room and selected profile, with visible confirmation and error handling. The exact create-versus-update interaction will be finalized with persistence rules.
- **Edit Room:** reopen all three guided steps prefilled, preserving values.
- **Browse/choose alternatives:** details page then Room Lab equipment selection; preserving working data.
- **No suitable match:** prominent amber **"No suitable match in the current catalogue"**, calculated requirement, explanation and pathway to edit or browse; *never* label an undersized AC suitable. A portable-only failure should say portable options are insufficient and offer split choices if possible.
- **Missing power/tariff:** do not fabricate numbers; show unavailable state or prompt for assumption.
- **Unsaved scenario:** user can view results and explore without saving.

24.5 Animation guidance

Allow restrained Best Match entrance, small progress fill or number fade, and subtle save success feedback. No confetti, spinning ratings or repeating show animations. The UI must remain usable with animations off.

25. Version 1.1 Change Log and Remaining Work

25.1 Changes from v1.0

- **Approved Modern Climate-Tech light theme** and practical component sizing and palette.
- **Finalized Home Dashboard visual/interaction design** with brand, hero room illustration, primary new-estimate action, offline/three-step caption, library/saved shortcuts, local cooling tip, conditional recent-room resume card, info panel and three-item bottom navigation.
- **Finalized three-step Room Setup visual design**, step navigation and validation, contextual summaries, illustration scope and **custom room name preservation** when room type changes.

- **Finalized Recommendation screen visual design:** dominant Best Match, capacity comparison bar with suitability states, two energy cards, dynamic explanation, alternatives, expandable breakdown, actions and no-match states.
- Added explicit instructions to avoid arbitrary energy figures, Home dummy metrics or physically exact room graphics.
- **No change** to the pre-existing simplified engineering heuristic, catalogue structure, scope or project deadlines.

25.2 Still pending after v1.1

1. **Room Lab detailed visual design:** hybrid workspace, selected/recommended comparison, accordion interactions, visual comfort card, reset/apply/save affordances.
2. **AC Library, AC Details, Saved Rooms detailed visual designs** and exact UI states.
3. **Dataset finalization:** all 14 complete records with powers, tiers, tags and explanations.
4. **Final numeric/logic lock:** cooling coefficients and bounds, portable rating convention, energy 0.65 factor and default LKR/kWh, simplified comfort thresholds.
5. **Technical architecture:** library choice, persistence strategy, saved-room lifecycle, route state transfer, error recovery, Android/offline test setup.
6. **Final Codex-ready PRD, test fixtures and implementation prompt** after the preceding points are approved.

Version status: Approved UI additions are incorporated, while listed unresolved choices stay explicitly provisional. Keep this file as the planning source of truth; update only when requested by the student.

Historical v1.1 snapshot ends here. Subsequent approved decisions in Sections 26–32 supersede earlier provisional or pending labels.

---

26. Version 1.2 Authority and Decision Status

Effective status: This section and Sections 27–32 contain approved decisions made after v1.1. Where historical sections 1–25 say *provisional*, *pending*, *not finalized*, or suggest Expo Router, the later v1.2 decisions take precedence. This is an additive update rather than a rewrite of previous context. The user explicitly approved each main UI design and Stage 6's numerical values, and instructed selection of the most suitable course-compatible technology.

26.1 Decision summary

Area	Current authoritative decision
Screens	Seven routes: Home, Room Setup, Recommendation, Room Lab, AC Library, AC Details, Saved Rooms
Room Lab layout	Hybrid dashboard; top room illustration and exactly three headline summary metrics; four accordions, only one expanded at a time
AC Library	Searchable/filterable 14-record catalogue; browse or selection mode
AC Details	Context-sensitive action: Back to Library, Use This AC, or Explore This AC in Room Lab
Saved Rooms	Open/Edit/Delete, delete confirmation, empty state, persistent records

Area	Current authoritative decision
Cooling coefficients	700 BTU/h/m² plus approved room/sun/window/roof/height/ occupancy rules in Section 6
Portable capacity	80% illustrative comparison capacity for 9K/12K portable profiles
Energy	30 days/month; fixed 0.65 average power-use multiplier; adjustable Rs. 40/kWh demonstration rate
Comfort	Qualitative Cool/Comfortable/Warm based on approved educational index (Section 28.4)
Framework	Expo + React Native + TypeScript
Navigation	**React Navigation** (Native Stack + Bottom Tabs), **not Expo Router**
User persistence	AsyncStorage; bundled AC profiles in local TypeScript/JSON
State	React Context and hooks; draft, temporary scenario, saved records remain separate
Illustration	`react-native-svg`, bundled visuals, no remote assets
Testing	Android device/emulator plus installed-build offline verification

26.2 Source/course alignment

The lecturer's two-page ICA 03 brief mandates **React Native**, 5+ screens, navigation, original local data, dynamic retrieval, search/filter/detail capability, completely offline operation, and individual GitHub repository with regular genuine commits. It does **not** mandate Expo/CLI or a specific navigation package. Available course-practical summaries mention React Navigation and typed route parameters, so React Navigation is a defensible course-aligned choice. The full original VLE lab archive was **not independently inventoried**; if the lecturer later provides a contradictory explicit setup constraint, honor it. Expo tooling is not synonymous with Expo Router: Expo can be used with React Navigation.

27. Approved UI Designs — Room Lab, Catalogue, Details, Saved Rooms

27.1 Room Lab: hybrid dashboard, approved

Purpose: Optional exploration after Recommendation, not the primary destination or a real physics simulator.

Top summary: Room name and room type, small responsive pseudo-isometric 2D room graphic, a current scenario/original/modified label, plus **three** compact live values: (1) estimated required BTU/h, (2) currently selected AC and/or capacity, and (3) approximate monthly energy *cost*. Do not add a large sticky illustration, extra metric rows or real-time sensor claims. The full kWh and explanatory details live within their respective sections. Illustration is bundled lightweight vector geometry, not a proportional floor plan.

Four expandable sections, accordion behavior:

1. **Room Conditions** — length, width, height (with valid unit handling), occupants, sunlight, window category, roof/ceiling category. Interactive changes immediately update the estimate and the current recommended profile; updates to illustrations are limited to simple sunlight/window/roof/occupant/type cues. For a quick-edit demonstration, a slider may represent area, but production must preserve coherent length/width dimensions and provide exact editable input fields; do not save area independently from dimensions.
2. **AC Selection** — show current automatic recommendation separately from the manually selected unit, capacity suitability and profile details. **Browse AC Library** launches the shared catalogue in selection mode; **Use This AC** returns to the same Room Lab scenario. Add small **Use Recommended AC** action to

restore automatic choice. Switching the equipment updates selected unit power, kWh, monthly charge and suitability but **never silently changes the calculated recommendedACId**.

3. **Energy & Cost** — daily hours (1–16), assumed LKR/kWh rate (starting Rs.40, suggested slider Rs.10–80 and optional custom numeric entry), monthly energy consumption kWh and illustrative additional monthly charge. Show assumption labels (30 days, 0.65 fixed factor). The manual selected AC is what drives the currently tested cost; where appropriate, compare its cost with the recommended AC. Do not recalculate fixed equipment input power from room condition changes.
4. **Comfort Explorer** — hypothetical temperature 20–34°C, RH 30–90%, and Still/Gentle/Strong airflow. Render only the educational label Cool, Comfortable or Warm, with simple color spectrum, contextual description and airflow artwork. No actual measured temperature or physical cool-down model; no PMV/PPD or numeric feels-like claim. Comfort controls do not modify electrical profile ratings or cooling demand.

Accordion: Only one expanded section at once by default; collapsed cards show readable labels and small helper text. Retain state while switching sections. Do not reinitialize the entire workspace when toggling an accordion.

Illustration fidelity: Reuse coherent SVG style between Home, Room Setup and Room Lab; variations may indicate bedroom/living/office arrangement, sun intensity, occupant count, window category, roof shape and airflow lines. No proportional layout engine, 3D model, AR, animated heat particles or furniture editor.

Actions: `Reset` reverts the temporary scenario to the entry baseline; `Apply Changes` updates the active working draft and navigates to the refreshed Recommendation; `Save Solution` remains on Recommendation, not an extra prominent Room Lab action. Attempting to leave a modified scenario prompts **Discard changes and leave Room Lab?**. A tested AC must remain explicitly distinguished from recommended equipment.

27.2 AC Library: approved

Two entry modes: Browse via bottom tab/Home; **Select** via Room Lab. Single shared catalogue UI, original bundled data, records rendered dynamically from list, not individually hand-built cards. Selected state or current tested equipment may be indicated in select mode.

Layout: Title **AC Library**, short offline-local subtitle, search bar (name/capacity/category/technology/keywords), category chips **All / Split / Portable**, capacity selector **All / 9K / 12K / 18K / 24K**, compact advanced filters for technology and efficiency, clear-filters action, live count **N of 14 profiles**, scrollable cards, friendly empty-results state. Search/filter changes react immediately; combine all active filters; clear restores full catalogue.

Catalogue card: Generic vector category icon (or locally bundled asset), name, nominal BTU/h, split/portable, technology and cost/efficiency tier; tap to navigate to AC Details. Capacity sort is useful; **no invented exact retail-price sorting, Buy Now button or retailer links**. Where the selected unit is insufficient for the room, show a label rather than censoring browsing results.

Empty state: Search icon, **No matching AC profiles**, explanatory text, **Clear Filters** action.

27.3 AC Details: approved

Structure: Profile name and efficiency/category badge, local split/portable illustration, nominal BTU/h, comparison BTU/h (explicitly separate for portable), input electrical power in kW, purchase tier, efficiency tier, profile description, distinctive features, best-use text, installation note and limitations.

Navigation action by entry context:

- Home/independent library browsing → show **Back to Library** (ordinary navigation back is also available).
- Room Lab → library selection → details → **Use This AC** sets selectedACId in active scenario and returns to Room Lab without losing edits.
- Recommendation alternatives → details → **Explore This AC in Room Lab** opens a new temporary scenario from the current working recommendation, then selects the viewed unit for comparison.

Handle unknown/missing record IDs with a friendly message and path back; do not fabricate a fallback profile. Generic catalogue is educational, not a shopping portal.

27.4 Saved Rooms: approved

Layout: Header **Saved Rooms**, subtitle *Your previous cooling solutions*, clear count, responsive card list, room-type icon, room name/type, m², last approximate BTU/h, recommended AC name and updated date. Primary actions **Open**, **Edit**, **Delete**; accidental deletion guarded by modal/confirmation.

Empty state: No saved rooms yet and **Create Room** button. No folders, tags, cloud sharing, PDF export or complex history.

Action behavior:

- Open → load persisted configuration into active draft, recompute with current heuristic/catalogue, navigate to Recommendation.
- Edit → prefill all three Room Setup steps without saving automatically.
- Find My AC after edit → recompute and show revised Recommendation; stored original remains unchanged until **Save Solution**.
- Save Solution on edited record → update the same ID; on new draft → create a new ID. Do not generate duplicate records accidentally.
- Delete → confirm before local removal; on failure preserve visible record and show error.

27.5 UI consistency

Maintain the approved Modern Climate-Tech light style from Sections 21–24: navy type (#0F172A), teal (#0D9488), cyan (#38BDF8), pale cyan (#E0F2FE), main off-white (#F8FAFC) and amber (#F59E0B) for warmth/caution; local vector illustrations; 16px-ish standard card radius, 24px hero, roughly 20px page padding, ~52px major buttons. Reusable AC cards, room summary cards, suitability bars, metric cards, chips, accordions, empty states and confirmations. Minimize repeated animations. The primary recommendation must remain visually dominant in the overall flow.

28. Approved Dataset, Numeric Rules and Example Fixtures

28.1 Complete 14-profile numeric table (approved; fictional educational profiles)

Stable ID	Display name	Nominal BTU/h	Comparison BTU/h	Electrical input kW	Purchase tier	Technology
SPL-09-STD	9K Standard	9,000	9,000	0.98	Budget	Fixed-speed split
SPL-09-INV	9K Inverter	9,000	9,000	0.78	Mid-range	Inverter split
SPL-09-ECO	9K Eco Inverter	9,000	9,000	0.69	Premium	Eco inverter split
SPL-12-STD	12K Standard	12,000	12,000	1.25	Budget	Fixed-speed split
SPL-12-INV	12K Inverter	12,000	12,000	1.05	Mid-range	Inverter split
SPL-12-ECO	12K Eco Inverter	12,000	12,000	0.90	Premium	Eco inverter split
SPL-18-STD	18K Standard	18,000	18,000	1.90	Budget	Fixed-speed split
SPL-18-INV	18K Inverter	18,000	18,000	1.55	Mid-range	Inverter split
SPL-18-ECO	18K Eco Inverter	18,000	18,000	1.35	Premium	Eco inverter split
SPL-24-STD	24K Standard	24,000	24,000	2.55	Budget	Fixed-speed split

Stable ID	Display name	Nominal BTU/h	Comparison BTU/h	Electrical input kW	Purchase tier	Technology
SPL-24-INV	24K Inverter	24,000	24,000	2.10	Mid-range	Inverter split
SPL-24-ECO	24K Eco Inverter	24,000	24,000	1.85	Premium	Eco inverter split
POR-09-CMP	Portable Compact	9,000	7,200	1.05	Budget	Portable
POR-12-PLUS	Portable Plus	12,000	9,600	1.40	Mid-range	Portable

These input powers and cost tiers were approved in Stage 6. **They are illustrative, fictional values, not product ratings.** Split comparison capacities use the nominal amount; portable comparison capacities use the explicitly approved illustrative factor of **0.80**, not an official certified rating. Include `category`, `technology`, `efficiencyTier`, `features[]`, `description`, `recommendedUse`, `installationNotes`, and `limitations` as originally defined; generate credible, distinct **generic** descriptions without invented manufacturer provenance. Portable units require an outdoor exhaust solution; evaporative coolers and window ACs remain excluded.

For the detail display, `coolingCapacityKW = nominalCapacityBTU × 0.000293071` (cooling kW, *not* electrical-input kW). Profile IDs shown here supersede abbreviated demo-only names and must be used consistently in references and saved records.

28.2 Cooling coefficients — approved final v1 implementation values

Retain the documented Section 6 approach exactly. Restated authoritatively:

```
area_m2 = length_m * width_m
base_BTUh = area_m2 * 700
factor = roomTypeFactor + sunlightFactor + windowFactor + roofFactor + (height_m - 2.7)*0.03
occupantExtra_BTUh = max(0, occupants - 2)*500
raw_BTUh = base_BTUh*(1 + factor) + occupantExtra_BTUh
estimated_BTUh = round(raw_BTUh/100)*100
```

Dimension	Values and factors
Room type	Bedroom 0%; Living +4%; Office +8%
Sunlight	Low −5%; Medium 0%; High +10%
Window	None −3%; Small 0%; Large +8%
Roof	Occupied floor above 0%; With ceiling +3%; Concrete slab +8%; No ceiling +12%
Ceiling height	Reference 2.7 m; change 3% per meter relative to reference
Occupants	First two baseline; +500 BTU/h for each extra person
Rounded output	Nearest 100 BTU/h

Default fixture: Bedroom 4 m × 3 m × 2.7 m, occupants 2, medium sun, small windows, roof with ceiling => 8,652 raw => **8,700 BTU/h**, suitable nominal class **9,000 split BTU/h**, preference Either/Balanced => **SPL-09-INV**. Small office/living room and high-load boundary tests must use the same deterministic engine. Metric/imperial conversion is a display/storage boundary, not an alternative formula.

28.3 Energy and cost — approved final v1 defaults

```
monthlyKWh = profile.illustrativeInputPowerKW * dailyUsageHours * 30 * 0.65
additionalEnergyChargeLKR = monthlyKWh * rateLKRPerKWh
```

- Default daily AC use = **8 hours/day** (range 1–16).

- Default number of usage days = **30** (fixed in v1).
- Average electrical-use multiplier = **0.65** (fixed in v1, not a power-factor correction).
- Initial rate = **Rs. 40/kWh**, deliberately a **demo assumption**, not a utility tariff assertion.
- Room Lab rate control: suggested **Rs. 10–80/kWh** with step 5; optional custom numeric field. Validate any custom entry and avoid negative/NaN values.
- UI label: **Estimated monthly AC energy cost** / **Approximate additional energy charge**; do not call it an exact before/after official Sri Lankan household bill.
- Calculated energy values should use actual selected (tested) AC profile. The Recommendation uses the auto-recommended profile. Do not apply changes in cooling load as hidden multipliers to power inputs.

Approved example: `SPL-12-INV` 1.05 kW, 8 h/day, 30 days, 0.65 → **163.8 kWh**; × Rs.40 → **Rs.6,552**. For the default bedroom `SPL-09-INV` 0.78 kW, 8 h/day → **121.68 kWh**; × Rs.40 → **Rs.4,867.20** (show appropriate display rounding). These are example calculations, not predictions of actual household billing.

28.4 Comfort Explorer rules — approved educational heuristic

This is not a measured or scientifically validated “feels like” temperature. Keep the calculated index **internal** and show only Cool/Comfortable/Warm plus a plain explanation and color bar.

```
index = selectedIndoorTemperatureC + humidityAdjustment - airflowAdjustment

Humidity adjustment:
  RH < 50%       -> 0
  50% <= RH < 70% -> +0.5
  RH >= 70%      -> +1.5

Airflow adjustment:
  Still  -> 0
  Gentle -> 0.5
  Strong -> 1.0

Label:
  index < 24         -> Cool
  24 <= index <= 28  -> Comfortable
  index > 28         -> Warm
```

Supported controls: 20–34°C temperature (default 26°C), 30–90% RH (default 60%), Still/Gentle/Strong airflow (default Gentle). Example: 26°C, RH60%, Gentle gives 26+0.5−0.5=26, **Comfortable**. At 29°C, RH80%, Still: 30.5, **Warm**. Comfort control updates must not change room load or electrical use. Add explicit educational disclaimer.

28.5 Remaining numeric display rules

- For an insufficient capacity, show amber **Below Estimated Requirement**; never “Suitable.”
- For nominal split capacity selection, the smallest capacity class satisfying `comparisonCapacityBTU >= estimated_BTUh` wins. Then use Standard/Inverter/Eco according to cost priority.
- Installation preference **Portable** with no sufficient portable model => explicit warning and optional split alternatives; avoid silently violating preference.
- Above largest sufficient catalogue comparison capacity => **No suitable match in current catalogue**.
- For markedly oversized manually selected equipment, show neutral **Higher Than Estimated Need** (exact UI warning threshold can be a presentation-only choice, but must not alter recommendation algorithm).
- We have deliberately *not* added a professional safety multiplier or real equipment cycle simulation.

29. Selected Technology, Navigation, Shared State and Offline Storage

29.1 Final stack (course compatibility considered)

Concern	Decision
Application	**React Native with Expo**
Language	**TypeScript**
Navigation	**React Navigation: Native Stack + Bottom Tabs**
Reference dataset	Bundled `acProfiles.ts` or JSON; fully local, no API
User data	**AsyncStorage**, using JSON serialization and versioned records
Shared state	React Context + hooks; no Redux
Calculation modules	Pure typed functions independent of view code
Styling	React Native StyleSheet + shared theme tokens
Room illustrations	`react-native-svg` and/or bundled local vectors
Icons	Locally bundled Expo-compatible icon resources
Remote services	None at runtime; no server, Firebase, Supabase, cloud accounts

Why not Expo Router? The lecturer brief does not require it; available course examples explicitly mention **React Navigation and TypeScript route typing**. React Navigation is the better course-aligned choice, while Expo retains a fast development workflow. A past-paper example is contextual, not a definitive lecturer toolchain mandate; original VLE practical ZIP has not been fully checked. Honor any later explicit restriction. Use package versions compatible with the chosen Expo SDK; avoid guessing or pinning obsolete major versions in this planning document.

Suggested layout in a standard Expo project (conceptual, not production code):

```
src/
  navigation/ RootNavigator.tsx, MainTabs.tsx, routeTypes.ts
  screens/ HomeScreen.tsx, RoomSetupScreen.tsx,
           RecommendationScreen.tsx, RoomLabScreen.tsx,
           ACLibraryScreen.tsx, ACDetailsScreen.tsx,
           SavedRoomsScreen.tsx
  components/ common/, room/, ac/, charts/
  data/ acProfiles.ts, coolingFactors.ts, comfortRules.ts
  types/ room.ts, ac.ts, navigation.ts
  context/ RoomContext.tsx
  services/ coolingEstimator.ts, recommendationEngine.ts,
            energyCalculator.ts, comfortCalculator.ts, storageService.ts
  theme/ colors.ts, spacing.ts, typography.ts
  assets/ images/
  utils/ units.ts, validation.ts
```

Correction: The earlier Expo Router `app/(tabs)/_layout.tsx` file-tree example in the historical section is **superseded**. Use a normal React Navigation `NavigationContainer` with typed Native Stack and Bottom Tab navigators; do not mix two routers.

29.2 Navigation contract

- Main bottom tabs: **Home** / **AC Library** / **Saved Rooms**.
- Stack screens: **Room Setup** / **Recommendation** / **Room Lab** / **AC Details**. Bottom tabs can be hidden on focused estimation screens.

- The AC Library is one shared route/component in independent browse mode or temporary equipment-selection mode.
- Pass lightweight IDs, origin/mode and context in typed navigation params; store the actual current room and scenario in shared React Context, not copied into every route param.
- Selecting from AC Details in selection mode updates the *active scenario's* `selectedACId` and returns to the existing Room Lab screen. Do not lose state by recreating Room Lab or blindly replacing the navigation stack.
- Detail screens opened from Recommendation alternatives can start Room Lab with the viewed AC selected while preserving the recommendation baseline.

29.3 Working-state separation — required

1. **RoomDraft** (session): the user's current editable room and preferences, including optional `sourceSavedRoomId`.
2. **RoomLabScenario** (temporary): copy of RoomDraft plus selected AC, energy rate and comfort controls; maintains immutable **entry baseline** for Reset.
3. **SavedRoomRecord** (persistent): AsyncStorage record changed only after successful Save Solution.

Automatic `recommendedACId` is **derived** each time from RoomDraft/Scenario and local AC dataset. `selectedACId` is user-chosen for exploration and must not automatically replace the recommendation. Users may explicitly choose **Use Recommended AC**.

Draft lifecycle: New Estimate creates default draft with no saved-record ID; loaded saved room carries its ID; editing does not modify saved data; `Find My AC` calculates but does not persist; save creates or updates according to source ID; after successful first save, associate working draft with the created ID to avoid repeated duplicate saves. A failed save must retain draft and show error.

Room Lab lifecycle: Begin from current draft/recommendation; initialize selected AC with recommended profile unless a user choice is being explicitly carried into exploration. Adjustments only modify scenario; Reset restores entry snapshot; Apply Changes copies scenario's supported room/preference/selected AC fields into working draft then returns to Recommendation; Save remains on Recommendation. Leaving changed Room Lab without applying presents a discard confirmation. Do not overwrite saved records during exploration.

Saved-room behavior: Open loads saved record, recomputes recommendation under current v1 model and shows Recommendation; Edit prepopulates three-step setup; Delete confirms then persists removal; all CRUD errors provide feedback and leave data safe. `sourceSavedRoomId` prevents accidental duplicate creation.

29.4 Persistence

Two data layers:

- **Read-only local AC records** bundled into the install package, searchable/filterable in memory; no network call.
- **AsyncStorage records** for saved rooms and small preferences (default unit, electricity rate). Use JSON serialization/deserialization, unique IDs, timestamps, `schemaVersion` and validation of parsed data. Treat corrupted data and missing referenced AC IDs gracefully; show a useful recovery/empty state rather than crash.

Suggested storage keys: `@acsize/savedRooms/v1` and `@acsize/settings/v1`. These names are implementation recommendations, not hard course requirements. Data should persist across app restarts; no server or login.

Suggested saved-room shape: `id`, `schemaVersion`, `name`, `roomType`, normalized `lengthM/widthM/heightM`, `preferredUnit`, `occupants`, `sunlight`, `windowCategory`, `roofConfiguration`, `installationPreference`, `dailyUsageHours`, `selectionPriority`, `selectedACId`, `electricityRateLKRPerKWh`, `lastEstimatedBTU`, `lastRecommendedACId`, `createdAt`, `updatedAt`. A recommendation snapshot is for quick display; active results should be recalculated when reopening. Comfort controls can remain temporary in v1 unless specifically saved as part of a scenario—do not infer actual indoor conditions.

29.5 Offline build/test and GitHub

- Use Expo during development, but offline requirement must be checked on an **installed bundled Android build**, not only Expo Go connected to Metro.
- Test all seven screens, local images/fonts/icons, catalogue search/filter/details, scenario swapping, calculation, save/reopen/edit/delete and restart in airplane mode.
- No online AC images, remote fonts, public APIs or cloud database. Development tooling may need internet to install dependencies, but app runtime should not.
- Maintain the student's own GitHub repo with genuine commits and pushes after milestones. No fabricated commits. Preserve complete source and all required local assets.

30. Consolidated Acceptance Scenarios for Codex

Test	Expected
Default room 4×3×2.7 m, bedroom, medium sun, small window, roof with ceiling, 2 people	Rounded **8,700 BTU/h**; Either/Balanced → **SPL-09-INV**
Change sunlight from medium to high, otherwise default	Cooling demand increases
Increase room size, otherwise same	Cooling demand increases
Change roof from with-ceiling to exposed	Cooling demand increases under heuristic
Increase occupants 2 → 3	Additional **500 BTU/h** before rounding
12K Inverter, 8 h/day, rate Rs.40	**163.8 kWh** and **Rs.6,552**
9K Inverter, 8 h/day, rate Rs.40	**121.68 kWh** and **Rs.4,867.20**
Increase hours, same AC/rate	kWh and energy charge rise
Raise tariff, same AC/hours	Charge rises; kWh unchanged
Comfort 26°C, RH60%, Gentle	**Comfortable**
Comfort 29°C, RH80%, Still	**Warm**
Portable-only, room needs 10,000 BTU/h	No portable match (max comparison **9,600**); explain and offer split option
Manual swap selected AC from recommendation	Original algorithmic recommendation preserved; selected energy calculation changes
Exit Room Lab after changes without Apply	Discard confirmation; saved record unchanged
Reset Room Lab	All scenario fields restored to entry baseline
Edit saved room, calculate, navigate back without Save	Original saved data unchanged
Save edited room	Update same ID, not duplicate
Open Library with filters leading to zero results	Friendly empty state and Clear Filters
Open unknown AC profile ID	Graceful unavailable state; never fabricate a profile
Installed Android build in airplane mode	All app core features work, including storage and local visuals

Additional UI verification: Seven distinct routes; three guided steps; bottom tabs; keyboard-safe fields; visually dominant recommendation card; Room Lab accordion (one open); capacity suitability text and color; local generic equipment illustrations; Save/Delete/Discard confirmations; empty/error states; responsive layout on small Android screen.

31. Version 1.2 Change Log

Added and approved after v1.1:

1. Detailed Room Lab visual/interaction specification: hybrid illustration, three headline metrics, four one-open accordions, local reactive vector effects, Reset/Apply, separate recommendation and tested selection.
2. Detailed AC Library UI: combined search and category/capacity/technology/efficiency filters, live count, browse/select modes, empty state.
3. Detailed AC Details UI: profile specs/notes and context-dependent selection/back actions.
4. Detailed Saved Rooms UI: cards, Open/Edit/Delete, empty state, confirmation and safe edit/save lifecycle.
5. All 14 fictional profile electrical-input ratings and cost tiers, plus portable comparison values.
6. Locked cooling heuristic values; fixed 0.65 factor and 30-day assumption; Rs.40/kWh editable demonstration rate.
7. Approved simple qualitative comfort index and thresholds.
8. Explicit recommendation/saved/selected AC distinction and state lifecycle.
9. Final stack: **Expo + TypeScript + React Navigation** Native Stack + Bottom Tabs; AsyncStorage, Context/hooks, local SVG and no backend. **Expo Router superseded.**
10. Expanded deterministic tests and offline installed-build criteria.

31.1 Remaining work (small, not new feature discovery)

- Final Codex implementation instructions and sequencing, including exact dependency versions compatible with the available Expo SDK.
- Finish unique, complete prose/tags/notes for each of 14 fictional profiles (numerical records are locked).
- Decide whether to expose optional custom electricity-rate text entry beyond the approved 10–80 slider (no change in calculation).
- Finalize microcopy and screen-level component placement during implementation without changing approved layouts.
- Establish Android test device/build method and verify availability of the instructor's actual lab tools if needed.
- The exact threshold for visually labelling an AC **Higher Than Estimated Need** is presentation-only and remains to be selected; capacity selection algorithm is already fixed.

Do not add new core screens, account systems, API dependencies, professional HVAC formulas or extensive simulations. The planning phase has intentionally locked the scope for the November 15, 2026 deadline.

32. Implementation Handover Guardrails

This document is now an implementation-oriented source of truth. When preparing the final Codex prompt, instruct Codex to:

- Read **all** of this specification, applying Sections **26–32** to supersede older pending/provisional statements.
- Use **React Navigation** (not Expo Router) and create a proper typed seven-route setup.
- Preserve *all* approved features including searching, filtering, details, local persistence, live calculations and interactive Room Lab.
- Respect fictional dataset and assumption labeling; never invent brand-specific retail pricing.
- Keep calculations in separate pure functions, not screen components.
- Implement incremental milestones with real Git commits and test each workflow before moving on.

- Explain main components, state lifecycle, dataset loading, formula and navigation in beginner-friendly terms after implementation.

Version 1.2 is a planning specification, not evidence that the app has been built or tested. The user's separate Codex development chat remains responsible for implementation and verification.

End of ACSize Master Specification v1.2 — updated, approved design and technical planning baseline.